SSGI

SPACE SCIENCE AND GEOSPATIAL INSTITUTE

FROM EARTH TO SPACE . . .

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Internship Management System – Full Documentation

1) Executive Summary

The Internship Management System (IMS) is a Django + PostgreSQL web application designed to automate and streamline the internship process at the **Space Science and Geospatial Institute (SSGI)**. It manages applications from students, requirement submissions by departments, admin approvals, and the matching process. The system ensures transparency, reduces manual effort, and centralizes data storage and reporting.

2) Problem Statement

Traditionally, internship applications and department requirements were handled manually (phone number, paper forms, spreadsheets). This approach introduced inefficiencies:

- **Scattered information** → Delays and miscommunication
- **Manual matching** → Time-consuming and error-prone
- **Opaque approvals** → Students unaware of application status
- **Tracking difficulties** → No consistent view of student progress or department capacity

IMS solves these by centralizing data and automating core workflows.

3) System Objectives

1. Provide a platform for students to apply online.
2. Allow departments to submit internship requirements digitally.
3. Enable admins to review, approve, and monitor applications.
4. Automate the matching process between students and departments.
5. Store data securely in PostgreSQL with export and reporting features.

4) How the System Works (Workflow)

1. **Department → Requirement Submission**
   Departments submit requirements (preferred fields, number of interns, required skills).

2. **Student → Application Form**
   Students complete and submit applications (CV, Internship Introduction Letter, passport ID, motivation letter).
3. **System → Database**
   Applications and requirements are stored in PostgreSQL.
4. **Database → Matching Algorithm**
   IMS runs a matching algorithm based on department needs and student skills/interests.
5. **Admin Review → Approval**
   Admins review matches, approve or reject applications, and assign interns.
6. **Approval → Notifications**
   Approved students are notified, and departments see assigned interns.
7. **Tracking & Reports**
   Admins monitor placements, export approved internships (Excel/CSV), and generate statistics via dashboards.

5) Features

- **Student Applications** – Online forms for internship submissions.
- **Department Management** – Requirement submissions with fields, preferred skills, and intern counts.
- **Approval Workflow** – Admin approval/rejection of applications and requirements.
- **Matching Algorithm** – Assigns interns based on skills and department needs.
- **Dashboard** – Provides real-time statistics on placements and pending approvals.
- **Database Integration** – Secure PostgreSQL backend.
- **Export Support** – Generate Excel/CSV reports.
- **Admin Panel** – Full management of users, departments, and matches.

6) System Architecture

- **Backend:** Django (Python)
- **Frontend:** HTML, CSS, JavaScript (Bootstrap)

- **Database:** PostgreSQL
- **Other Tools:** Django Admin Panel, openpyxl (Excel export)

7) Installation

1. Clone the repository

```
git clone https://github.com/your-username/internship-management-system.git

cd internship-management-system
```

2. Set up the Python environment

```
python -m venv venv

venv\Scripts\activate      # Windows

# or

source venv/bin/activate   # Mac/Linux
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Configure PostgreSQL

- Create a PostgreSQL database named intern.
- Update DATABASES in stellar_core/settings.py with your PostgreSQL credentials (or use .env for secrets).

5. Run migrations

```
python manage.py makemigrations

python manage.py migrate
```

6. Create a superuser

```
python manage.py createsuperuser
```

7. Start the development server

```
python manage.py runserver
```

8) Usage

• Admin Panel: Manage students, departments, matches, and approvals at 👉 http://127.0.0.1:8000/admin/

• Students: Apply for internships through the application form at /apply/.
• Departments: Submit requirements via the department portal at /departments/.

> ⚠️ Important: Department users cannot register themselves. They must first be created by the Admin in the Django Admin Panel and assigned to the department group. The department portal must also be set to OPEN for requirement submissions.

• Approvals: View and approve submitted applications and requirements in the admin panel.

8.1 STEPS TO REGISTER USERS FOR DEPARTMENT

Create a department user (step-by-step)

1) Open Django admin

Go to: http://127.0.0.1:8000/admin/

Log in with your superuser (admin) account.

2) Ensure the group exists

In the left menu, click Groups → Add Group.

Name: department (exact lowercase) → Save.

3) Create the user

Left menu → Users → Add user.

Enter Username (e.g., dep_it), Password, Confirm password → Save.

4) Assign group

After saving, you'll see the user's change page.

Groups → move department to Selected groups → Save.

Leave Staff status and Superuser status unchecked.

5) Open the portal (if closed)

Left menu → Departments portal config.

Set it to OPEN or use action “Open for 7 days” → Save.

6) Test login as department user

Visit: http://127.0.0.1:8000/accounts/login/

Log in with the new user.

You should be redirected to: http://127.0.0.1:8000/departments/

Notes:

Access is granted by group membership only (department). No extra permissions needed.

Admins can access everything; department users only see /departments/.

9) Project Structure

```
intern/
|— apps/
|   ├— accounts/        # Authentication & user accounts
|   ├— adminpanel/      # Custom admin dashboard
|   ├— applications/    # Student applications module
|   ├— approved/        # Approved internships
|   ├— departments/     # Department requirements
|   ├— matches/         # Matching logic
|   ├— progress/        # Progress tracking
|— motivation_letters/  # Uploaded motivation letters
|— passport_ids/        # Uploaded passport IDs
|— recommendations/     # Uploaded recommendation letters
|— resumes/             # Uploaded resumes
|— templates/           # HTML templates (students, depts, admin)
|— stellar_core/        # Core Django project settings
|— manage.py
```

```
|— requirements.txt
|— .env              # Secrets & environment variables
```

10) Database Models

- **InternshipApplication** – Student data (resume, motivation letter, GPA, ID, recommendations).
- **Department** – Requirements with preferred skills/fields and intern slots.
- **Match** – Stores match results and scores.
- **Approved** – Finalized internships with registration status.
- **ProgressView** – Tracking progress and generating statistics.

11) Matching Algorithm (Simplified)

- Compare student skills/interests with department requirements.
- Calculate scores based on overlap and eligibility.
- Assign interns to departments until capacity is filled.
- Admin can adjust or override assignments before approval.

12) Security & Access Control

- Role-based access: **Admin**, **Department**, **Student**
- Authentication via Django accounts
- Secure file upload handling (resumes, IDs, letters)
- PostgreSQL with role-based DB access

13) Deployment

- Local: python manage.py runserver
- Production: Use Nginx/Gunicorn + PostgreSQL

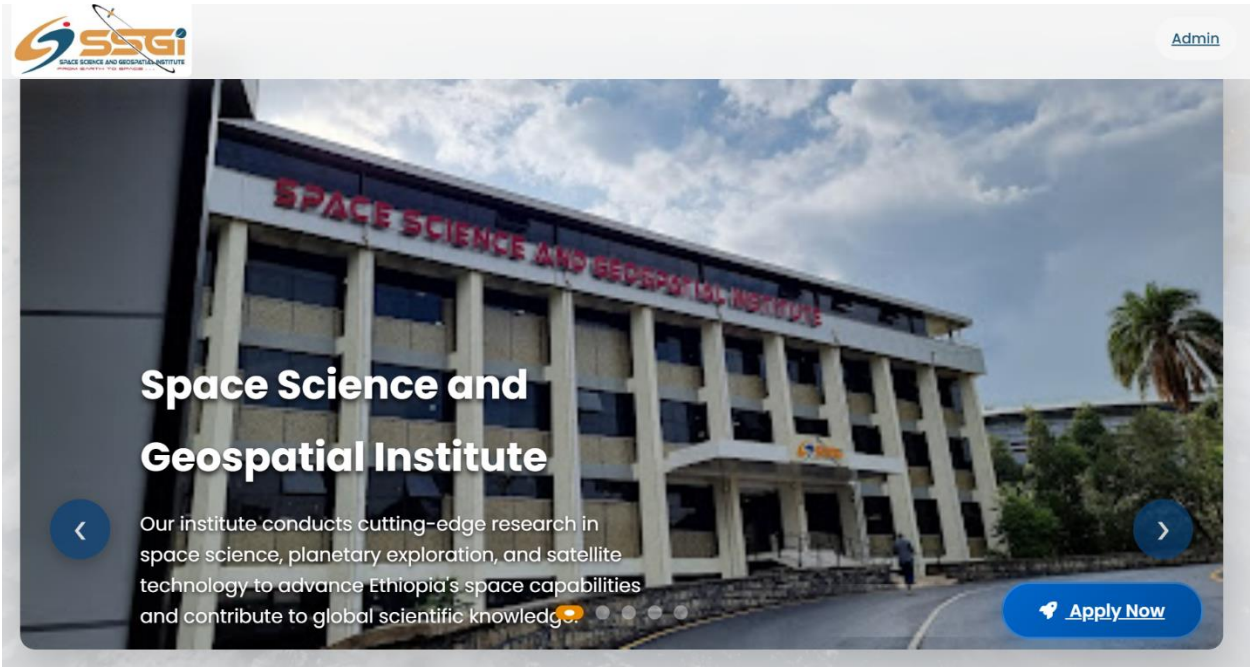
- Environment secrets stored in .env
- Daily DB backups recommended

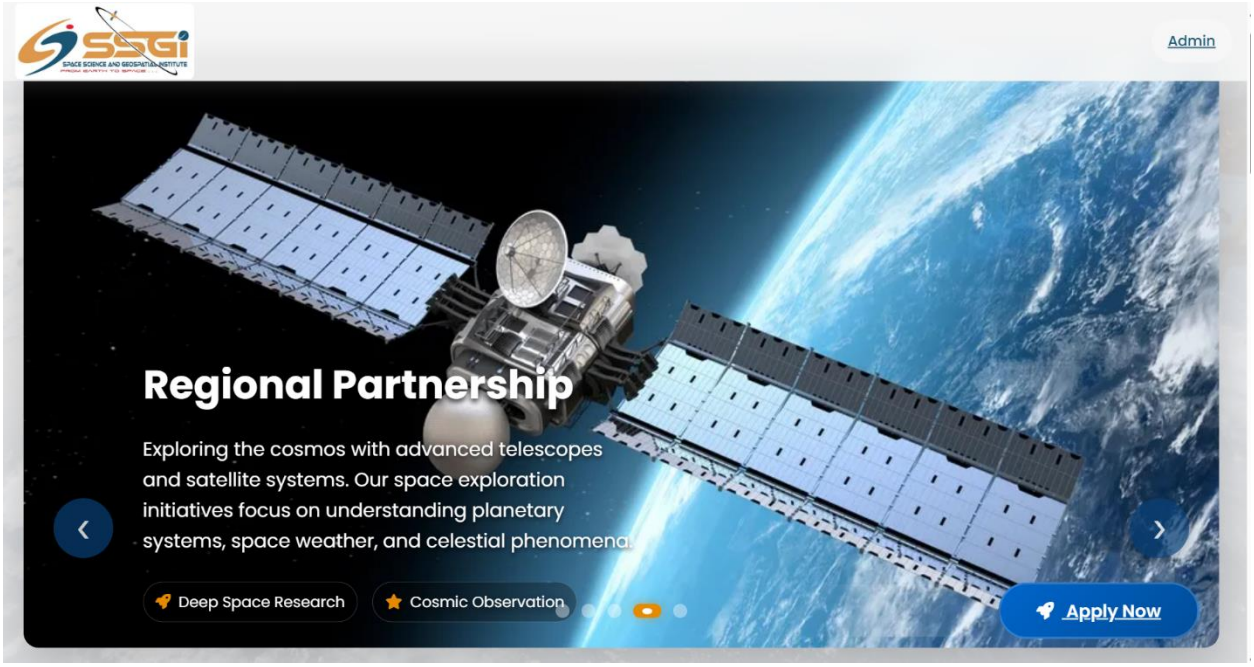
14) Benefits & Impact

- **Efficiency**: Automated matching and approvals reduce admin load.
- **Transparency**: Students and departments can view real-time status.
- **Scalability**: Supports more applicants and departments with minimal overhead.
- **Accountability**: Audit trail via Django Admin actions.
- **Reporting**: Export to Excel/CSV for tracking and external use.

15) Screenshots

HOME PAGE

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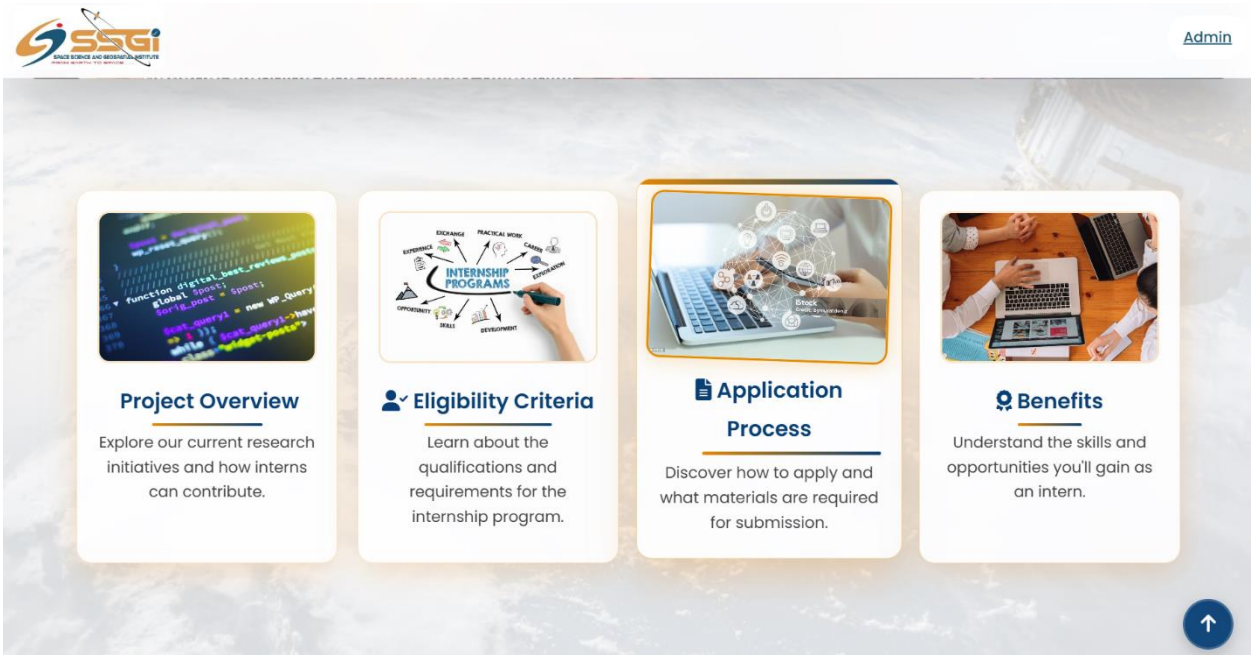
Regional Partnership
Exploring the cosmos with advanced telescopes and satellite systems. Our space exploration initiatives focus on understanding planetary systems, space weather, and celestial phenomena.
Deep Space Research
Cosmic Observation
Apply Now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Project Overview
Explore our current research initiatives and how interns can contribute.
INTERNSHIP PROGRAMS
Eligibility Criteria
Learn about the qualifications and requirements for the internship program.
Application Process
Discover how to apply and what materials are required for submission.
Benefits
Understand the skills and opportunities you'll gain as an intern.

STUDENT APPLICATION FORM

Home

SSGI INTERNSHIP PROGRAM

Join our mission to explore the cosmos and advance geospatial sciences. Gain hands-on experience with cutting-edge technology and work alongside leading scientists in space research.

Apply as Intern

Home

INTERNSHIP APPLICATION

Please fill out the form to apply for our internship program

Personal Intern

First Name*

Last Name*

Age*

Email Address*

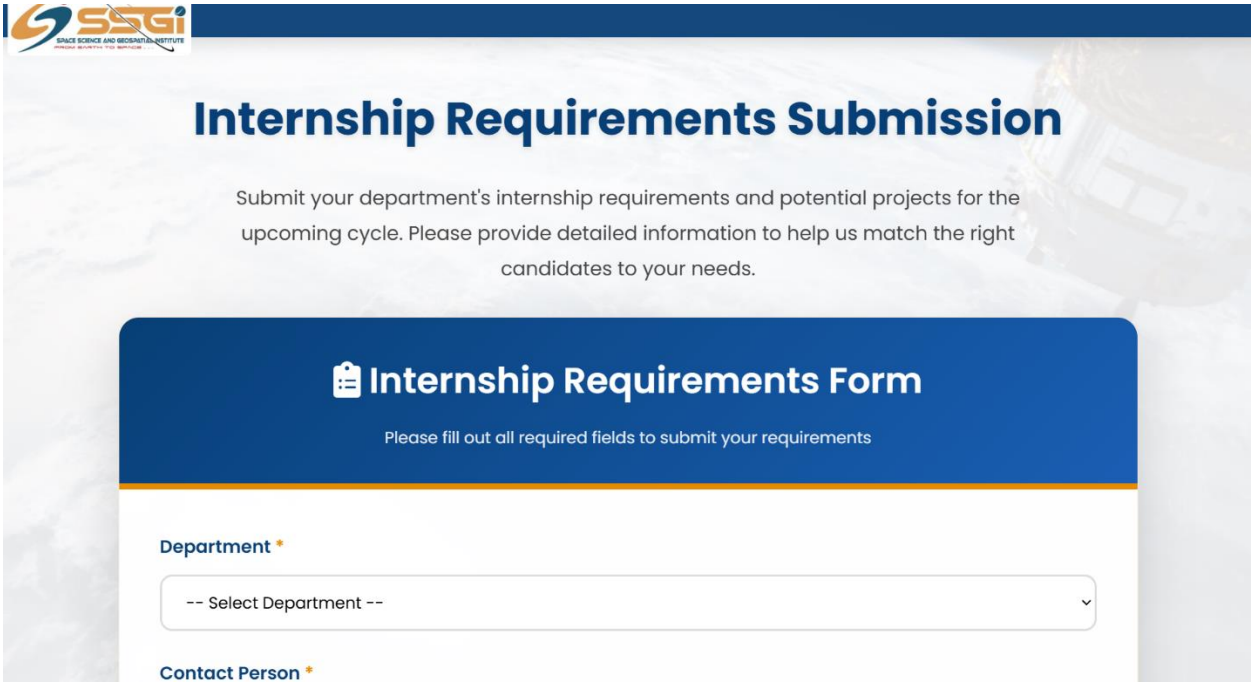
Phone Number*

City*

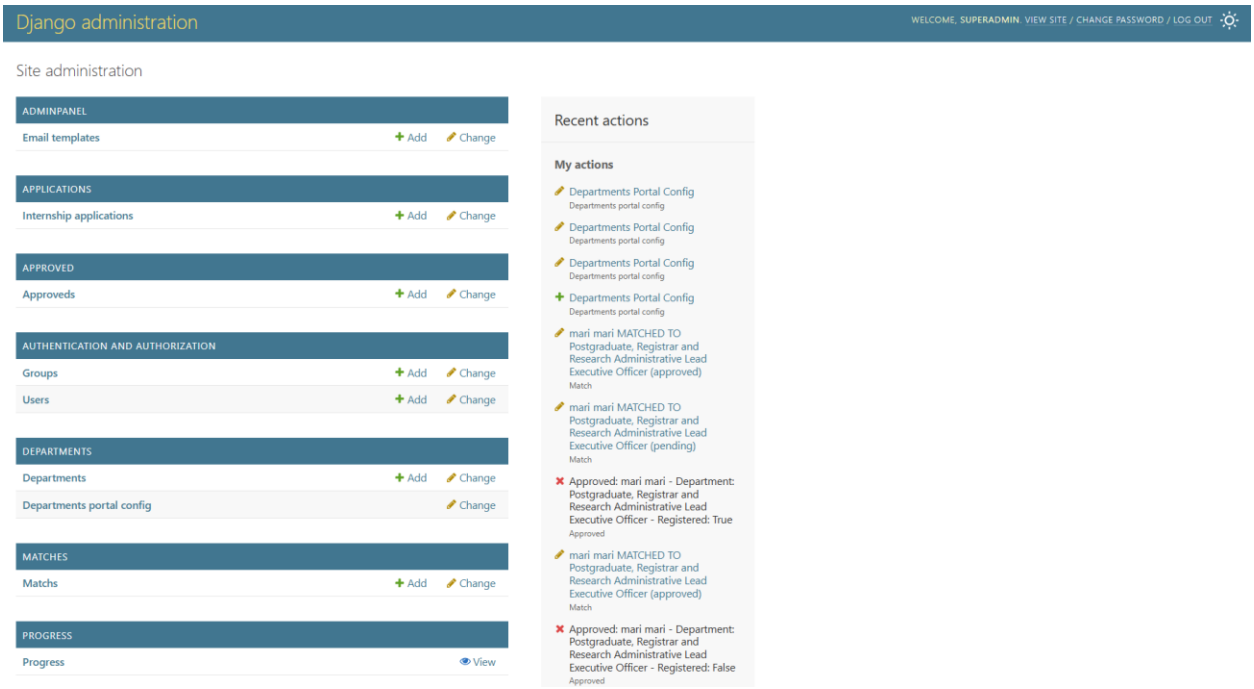
University (Ethiopia)*

If College, Enter Name

- **DEPARTMENT REQUIREMENT SUBMISSION**



ADMIN DASHBOARD



16) Conclusion

The Internship Management System provides a centralized, transparent, and efficient way for SSGI to handle internships. By automating application intake, department requirement submissions, approvals, and matching, it reduces manual errors, improves fairness, and enables data-driven decision-making.